

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



May 22, 2007

MATERIALS LABORATORY FACTUAL REPORT

Report No. 07-070

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : I-90 Connector Tunnel to Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. TESTS CONDUCTED

Creep testing of Powers Fasteners Power-Fast Fast Set epoxy per ICC-ES AC 58 and ASTM E 1512.

C. DETAILS OF THE TESTING

Powers Fasteners¹ Power-Fast Fast Set epoxy anchor system was sent to an outside laboratory² for testing to assess the creep behavior of anchors under sustained tension loading. The testing was performed in accordance with ICC Evaluation Service (ICC-ES) Acceptance Criteria AC 58³ and ASTM E 1512⁴. The report documenting the testing is attached.

The Safety Board provided a dual cartridge of Fast Set resin and hardener along with a manual injection tool and static mixer nozzles. The cartridge was marked with lot number 062890I1LC.

Creep testing per ICC-ES AC 58 and ASTM E 1512 is carried out on anchors consisting of 0.5-inch diameter threaded steel rod embedded to a depth of 4.5 inches. (The report incorrectly indicates that the anchors were 0.25 inch in diameter. A correction noting that the anchors were 0.5 inch in diameter is appended.) The test protocol has three phases. First, five anchors are pulled at room temperature to determine an average ultimate strength. Second, five anchors are pulled at a minimum of 110 °F in order to

¹ Powers Fasteners, Brewster, New York.

² Standard Testing and Engineering Company, Tulsa, Oklahoma.

³ Acceptance Criteria for Adhesive Anchors in Masonry Elements AC 58, ICC Evaluation Service, Whittier, California.

⁴ ASTM E 1512, Standard Test Methods for Testing Bond Performance of Bonded Anchors, ASTM International, West Conshohocken, Pennsylvania.

determine the average anchor displacement at the point of maximum load; this displacement is used as a gauge against which to measure the displacement of the anchors in the creep test. The creep testing itself is also to be performed at a minimum of 110 °F, using three anchors. The tensile force applied in the creep test is specified to be 40 percent of the average anchor ultimate strength determined from the tests performed at room temperature.

There was substantial scatter in the room temperature strength data, and the procedure given in ICC-ES AC 58 and ASTM E 1512 requires that strength measurements less than 85 percent of the average of the five tests be excluded and then the average strength recalculated. This procedure required three of the five tests to be discarded. The average of the five tests was 9,213 pounds, and the recalculated average was 14,089 pounds.

Powers Fasteners product literature lists the ultimate strength of 0.5-inch diameter threaded rod anchors installed to 4.5 inches embedment in 3,000 psi strength concrete using Power-Fast Standard Set epoxy as 13,600 pounds. (The June 2005 version of AC 58 specified that the concrete used have a compressive strength between 2,500 and 3,500 psi.) The table containing the Power-Fast Standard Set data has a footnote stating "Reduce the above ultimate load values by 25 percent when calculating ultimate load capacities for the Fast Set formula." Powers Fasteners product literature therefore indicates that the ultimate strength of these anchors installed with Fast Set should be 10,200 pounds.

Given the amount of variability leading to this result, it was concluded that the strength value of 10,200 pounds indicated in Powers design literature was more appropriate to use as a baseline for testing. The creep load applied at 110 °F was therefore 40 percent of 10,200 pounds or 4,080 pounds.

All three of the anchors tested at 110 °F at a tensile load of 4,080 pounds pulled out in less than 24 hours.

Carl R. Schultheisz
Materials Research Engineer



TULSA OFFICE

5358 S 125 East Ave Ste B • Tulsa, OK 74146-6208
(918) 459-2700 • (800) 725-4592 • (918) 459-2715 FAX

902 Trails West Loop
Enid, OK 73703-6336
(580) 237-3130
(800) 725-3130
(580) 237-2311 FAX

202 SE "J" Ave
Lawton, OK 73501-2481
(580) 353-0872
(800) 725-0872
(580) 353-1263 FAX

3400 N Lincoln Blvd
Oklahoma City, OK 73105-5493
(405) 528-0541
(800) 725-0541
(405) 528-0559 FAX

May 16, 2007

National Transportation Safety Board
NTSB RE-30
490 L'Enfant Plaza East, SW
Washington, DC
20594

Attention: Carl Schultheisz
Materials Research Engineer

Subject: Report on the Performance of Powers Fasteners' "High Strength Epoxy Anchoring System for Concrete and Masonry" Bolt Epoxy Power-Fast+ Fast Set Formula

Dear Mr. Schultheisz:

Standard Testing and Engineering Company (Standard Testing) is pleased to present this report of the results of the above mentioned test our company performed for the NTSB.

The attached report entitled "Standard Testing and Engineering Company's report on the performance of Powers Fasteners' 'High Strength Epoxy Anchoring System for Concrete and Masonry' bolt epoxy Power-Fast+ Fast Set Formula" presents the static tension test and creep test results of the aforementioned epoxy as outlined in ASTM E 1512-01 and ICC-ES AC58. The tests described in this report were conducted from March 4, 2007 to March 15, 2007 at Standard Testing and Engineering Company's "Pine Street" office in Tulsa, Oklahoma.

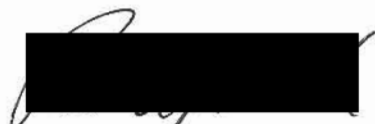
All of the tests described in this report were supervised by Randall Baker, a Professional Engineer licensed in the State of Oklahoma, and other Standard Testing personnel. Standard Testing is qualified to perform and supervise these tests as per ICC-ES License Number TL-187.

This report has been reviewed by Standard Testing for accuracy with respect to our testing and observations and is, to the best of our knowledge, an accurate representation of the conditions, procedures, and results of the specific tests described herein. This report is being issued by Standard Testing in conformance with the requirements of ASTM E 1512-01 and ICC-ES AC58. This report applies only to the standards or procedures indicated and to the test specimens observed and is not

necessarily indicative of the performance of apparently identical or similar products at another location. This report shall not be reproduced except in full, without the written approval of Standard Testing.

The Powers Fasteners' 'High Strength Epoxy Anchoring System for Concrete and Masonry' Power-Fast+ Fast Set Formula bolt epoxy was sampled in accordance with ASTM E 1512-01 and the manufacturer's written instructions, a copy of which is enclosed. The lot number for the epoxy is #06289011LC.

Respectfully submitted,
STANDARD TESTING AND ENGINEERING COMPANY



Randall Baker, P.E.
Senior Engineer



Miles Hunter, P.E.
Tulsa Operations Manager

(IAS listing #TL-187; Oklahoma CA#77 expires 6-30-07)

Attachments (12 pages) + copy of manufacturer's instructions



Test Report

Date: May 16, 2007

Standard Testing and Engineering Company's Report on the Performance of Powers Fasteners' 'High Strength Epoxy Anchoring System for Concrete and Masonry' Bolt Epoxy – Power-Fast+ Fast Set Formula

1.0 INTRODUCTION

This report presents the static tension test and creep test results of the aforementioned epoxy as outlined in ASTM E 1512-01 and ICC-ES AC58. The tests described in this report were conducted from 3/6/2007 to 3/15/2007 at Standard Testing and Engineering Company's "Pine Street" office located at 10838 E. Newton Street, Suite 113 in Tulsa, Oklahoma, 74116. Test results are summarized below and presented in complete detail in the appendix.

All tests were performed, witnessed, and verified by Standard Testing and Engineering Company (STEC), an independent, ICC-ES listed testing agency located at 5358 S. 125th E. Avenue, Suite B, Tulsa, Oklahoma 74146; telephone (918) 459-2700. The ICC-ES listing number for Standard Testing and Engineering Company is TL-187. This test was performed as per the direction from our client, the National Transportation Safety Board (NTSB).

2.0 PRODUCT DESCRIPTION

From Power-Fast product literature – "The Power-Fast+ Epoxy adhesive system is a two-component structural epoxy which is packaged in engineered plastic cartridges. It is used with either a manual, pneumatic or power-operated injection tool and proportionally mixed through a static-element mixing nozzle. Power-Fast+ is a premium, non-sag, high-strength epoxy which has been vigorously tested to meet or exceed required standards as an anchoring and bonding adhesive Power-Fast+ Epoxy is designed for use in anchoring threaded rods, bolts, reinforcing bars, and smooth dowels into concrete and masonry base materials. The system can also be used to anchor into hollow masonry materials using rod and rebar with screen tubes. In addition to anchoring applications, Power-Fast+ is used for bonding steel and cured concrete to cured concrete, for pick proofing applications, for surface sealing cracks and for placing injection ports."

The specific product used for this test was Powers Fasteners' 'High Strength Epoxy Anchoring System for Concrete and Masonry' bolt epoxy – Power-Fast+ Fast Set Formula (epoxy). (Photo 1)

Photo No. 1: Epoxy used in this test



3.0 EPOXY/BOLT INSTALLATION PROCEDURE

This epoxy was tested by using it to anchor sixteen (16) 8" long, 1/4" diameter, all-threaded (according to ANSI B1.1 UNC Series Class 2A) steel bolts into a 90 3/4" x 36" x 12" deep slab of 2710 psi concrete. The bolts were anchored 4 1/2" deep with the epoxy in accordance with ASTM E 1512-01 "Standard Test Methods for Testing Bond Performance of Bonded Anchors", the instructions from the NTSB, and also Powers' Fasteners recommended installation instructions. Tools used included a hammer drill, a nylon brush, an air compressor (to blow the holes clean), and a vacuum cleaner (to remove any remaining concrete dust).

Sixteen (16) holes were drilled into the concrete slab. The holes were all drilled a minimum of two embedment depths (9") away from any edge and any other hole, as per ASTM E 1512-01. Five holes were used for the Static Tension Test at 75°F, five holes were used for the Static Tension Test at 110°F, three holes were used for the Creep Test at 110°F, and three holes were drilled as spares. All of the holes were drilled to a depth of 4 1/2", as per ICC-ES AC 58. The holes were then blown clean with compressed air, cleaned with a nylon brush, and then blown clean again (as per the installation instructions from Powers' Fasteners). The holes were then all vacuumed with a shop vac to remove any remaining dust.

Next, as recommended by the manufacturer, a portion of the epoxy was pushed out as waste to ensure that proper mixing of the two components was taking place. Then, each hole was filled 2/3rds full with epoxy starting from the bottom of the hole. The application tube was slowly withdrawn as the epoxy was placed so as to avoid creating air pockets.

After each hole was filled with epoxy as stated above, an 8 inch long threaded bolt was pushed into the hole while being turned slightly to ensure positive distribution of the adhesive. It was ensured that the threaded bolt was free of dirt, grease, oil and any other foreign material prior to placement. Care was also taken to ensure that the bolt was fully seated at the bottom of the hole and that some adhesive had flowed from the top of the hole after installation, as per the Powers Fasteners' instructions.

All of the epoxy was then allowed to harden for a minimum of 24 hours, as per the recommended time stated on the product packaging.

4.0 TEST MATERIALS

4.1 Epoxy: The epoxy used is as described above and was installed and tested in accordance with ASTM E 1512, the manufacturer's instructions, and direction from the NTSB.

4.2 Concrete: All testing was conducted in uncracked limestone with only one #4 piece of re-bar in a "C-shape", laying parallel to the top and bottom surfaces, embedded 4 inches deep from 3 sides perpendicular to the top and bottom surfaces, and flush with the remaining side. Normal-weight, natural river sand and normal-weight crushed rock as commonly used by the Tulsa, Oklahoma concrete industry were used as aggregate meeting ASTM C 33-90. A 2,000 psi concrete mix design was used to obtain the required strength range. (Actual strength of concrete was 2710 psi – see break reports in the appendix).

Concrete cores were prepared and tested in accordance with ASTM C 42 and ASTM C 39-96, respectively. The concrete structural block member measured 90 3/4" long x 36" wide by 12" deep with test surfaces replicating a smooth trowel finish.

Concrete compressive strength verification reports are provided in the appendix.

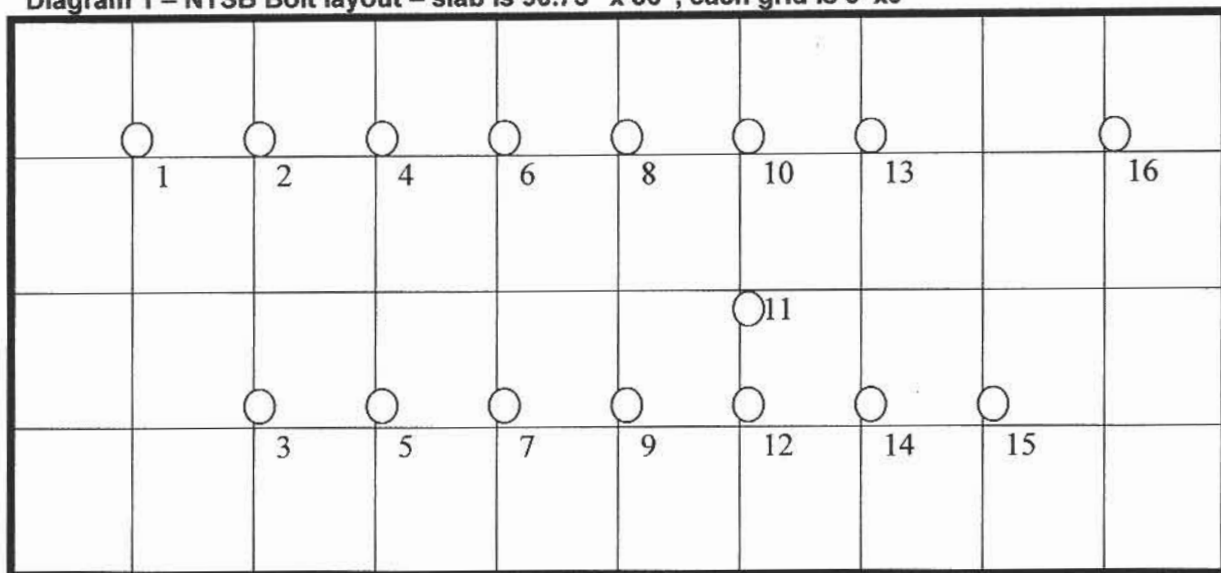
4.3 Anchor Bolts: The bolts used in testing are cut from uncoated, ASTM A 193 Grade B7 rod, threaded to ANSI B1.1 UNC Series Class 2A. The bolts were installed and tested in accordance with ASTM E 1512 and direction from the NTSB. The bolt diameter is 1/2 inch and the hole diameter is 9/16 inch.

5.0 TEST EQUIPMENT AND PROCEDURE

Testing was conducted in accordance with ASTM E 1512-01, "Standard Test Methods for Testing Bond Performance of Bonded Anchors".

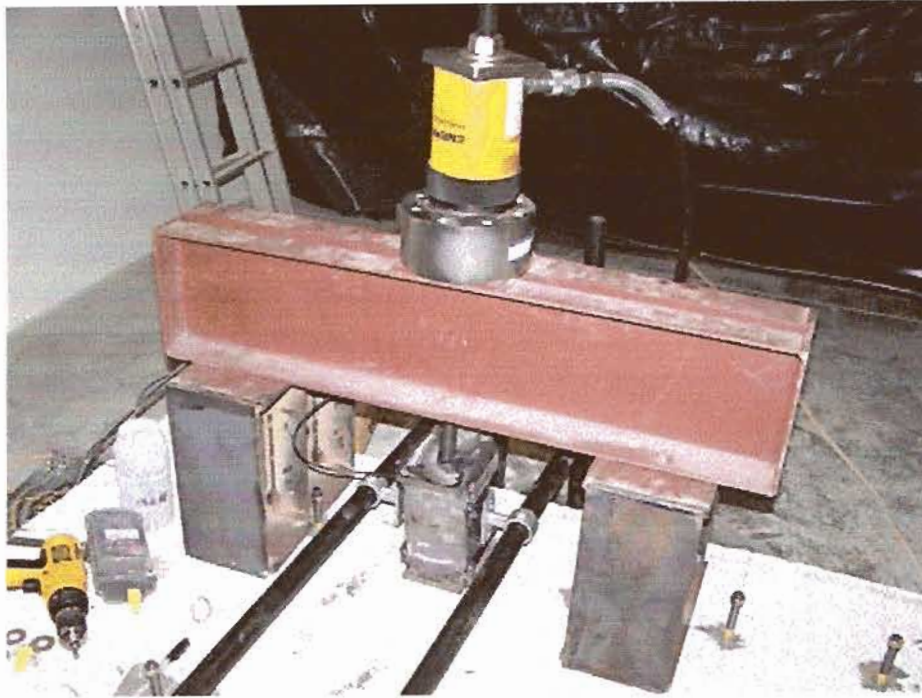
General – All bolts were tested after the minimum required 24 hour time period epoxy set time had elapsed. The epoxy and bolts were set on March 4, 2007 from 9:00 – 10:00 AM and the first bolt/epoxy testing began at 3:30 PM on March 6, 2007. A total of sixteen bolts were installed with the epoxy (5 for each Static Tension Test, 3 for the creep test, and 3 spares). Diagram 1 shows the numbering system used for the bolts.

Diagram 1 – NTSB Bolt layout – slab is 90.75" x 36", each grid is 9"x9"



Static Tension Test at 75°F +/- 10°F – Five bolts (as per ASTM E 1512-01) were used for this test. The bolts used were bolt numbers 1, 2, 4, 6 and 11 (see Diagram 1). A testing and measuring apparatus was assembled and used in accordance with ASTM E 1512-01. Load was applied and measured from the top of the beam via a jack and load cell system (see Photo 7). The load pulled up on a coupler that attached on the other end to the bolt being tested. Deflection was measured at two points on the coupler. These bolts were pulled from 3:30 pm to 6:00 pm on March 6, 2007. The amount of load required to pull the bolt out was measured, as per the ASTM. The results ranged from 4,575.9 pounds to 18,057.1 pounds. The average, calculated according to ASTM E 1512-01, is 14,088.7 pounds. Complete test results are included in the appendix.

Photo No. 7: Static Tension test on bolt #4



Static Tension Test at Min of 110°F – Five bolts (as per ASTM E 1512-01) were used for this test. The bolts used were bolt numbers 5, 7, 9, 16 and 8 (see Diagram 1). The same testing and measuring apparatus used for the 75°F test was used for this test. After heating the slab to a temperature of 110°F min, load was applied and measured from the top of the beam via a jack and load cell system. The load pulled up on a coupler that attached on the other end to the bolt being tested. Deflection was measured at two points on the coupler. These bolts were pulled on March 12, 2007. The amount of deflection that occurred until failure was measured, as per the ASTM. The results ranged from 0.06 inches to 0.25 inches. The average, calculated according to ASTM E 1512-01, is 0.10 inches. Complete test results are included in the appendix.

Photo No. 9: creep test apparatuses



Creep Test at min of 110°F – Three bolts (as per ASTM E 1512-01) were used for this test. The bolts used were bolt numbers 2, 13, and 15 (see Diagram 1). Three testing and measuring apparatuses were assembled and used in accordance with ASTM E 1512-01. Each apparatus consisted of a metal footing to secure one end of a vertical square tube in place, a long horizontal square tube joined at the other end of the vertical tube, and a threaded rod hanging freely from the horizontal arm on which weights can be placed and allowed to “creep” the bolt to failure via transference of the load to the bolt being tested. The amount of weight applied was 40% of the average ultimate load from the static tension test at 75°F +/- 10°F, as per ASTM E 1512-01. The average ultimate load at 75°F +/- 10°F was 14,089 pounds, so the load that should be applied in the creep tests is 5636 pounds. However, due to the wide scatter in the 75 degree static results, Standard Testing consulted with the NTSB and it was decided to use 40% of the ultimate load that the Powers’ literature says that a bolt of this diameter and embedment depth can handle (when using Fast Set epoxy in 3000 psi concrete), which works out to 4080 pounds. The weights were placed on our lever-arm weight apparatuses and the bolts were pulled from March 14 to March 15, 2007. Because the bolts experienced catastrophic failure within 24 hours of the start of the test (even before we had the temperature all the way to 110 degrees F), no usable data could be extrapolated.

TEST RESULTS

Static Tension Test at 75°F +/- 10°F

Bolt #	Max Load, pounds
1	10,120.3
2	4,575.9
4	6,284.9
6	7,028.9
11	18,057.1

The average load of the 75 °F static test is 9,213.42 pounds, and according to ASTM E 1512-01, any values less than 85% of the average are thrown out and the remaining values are re-averaged, which means that the

values for bolt #2, #4, and #6 were thrown out. When the remaining two (bolt #1 and #11) are averaged, the **result is 14,088.7 pounds**. According to the manufacturer's literature, the epoxy should be able to handle an applied tensile strength of 12,980 pounds for 2,000 psi concrete and an applied tensile strength of 13,600 pounds for 3,000 psi concrete. The manufacturer's product literature is included in the appendix. The average compressive strength from the concrete slab we used for this test is 2710 psi.

Static Tension Test at minimum of 110°F

Bolt #	Displacement, in.
5	0.24
7	0.25
8	0.14
9	0.10
16	0.06

The average displacement of the 110 °F (minimum) static test is 0.16 inches, and according to ASTM E 1512-01, any values greater than 115% of the average are thrown out and the remaining values are re-

averaged, which means that the values for bolt numbers 5 and 7 were thrown out. When the remaining three (bolts #8, #9, and #16) are averaged, the **result is 0.10 inches**.

Creep Test at minimum of 110°F

As summarized above, these bolts experienced catastrophic failure within 24 hours, so no usable data was collected.

Respectfully submitted:

STANDARD TESTING AND ENGINEERING COMPANY


Randall Baker, P.E.
Senior Engineer


Miles Hunter, P.E.
Tulsa Operations Manager



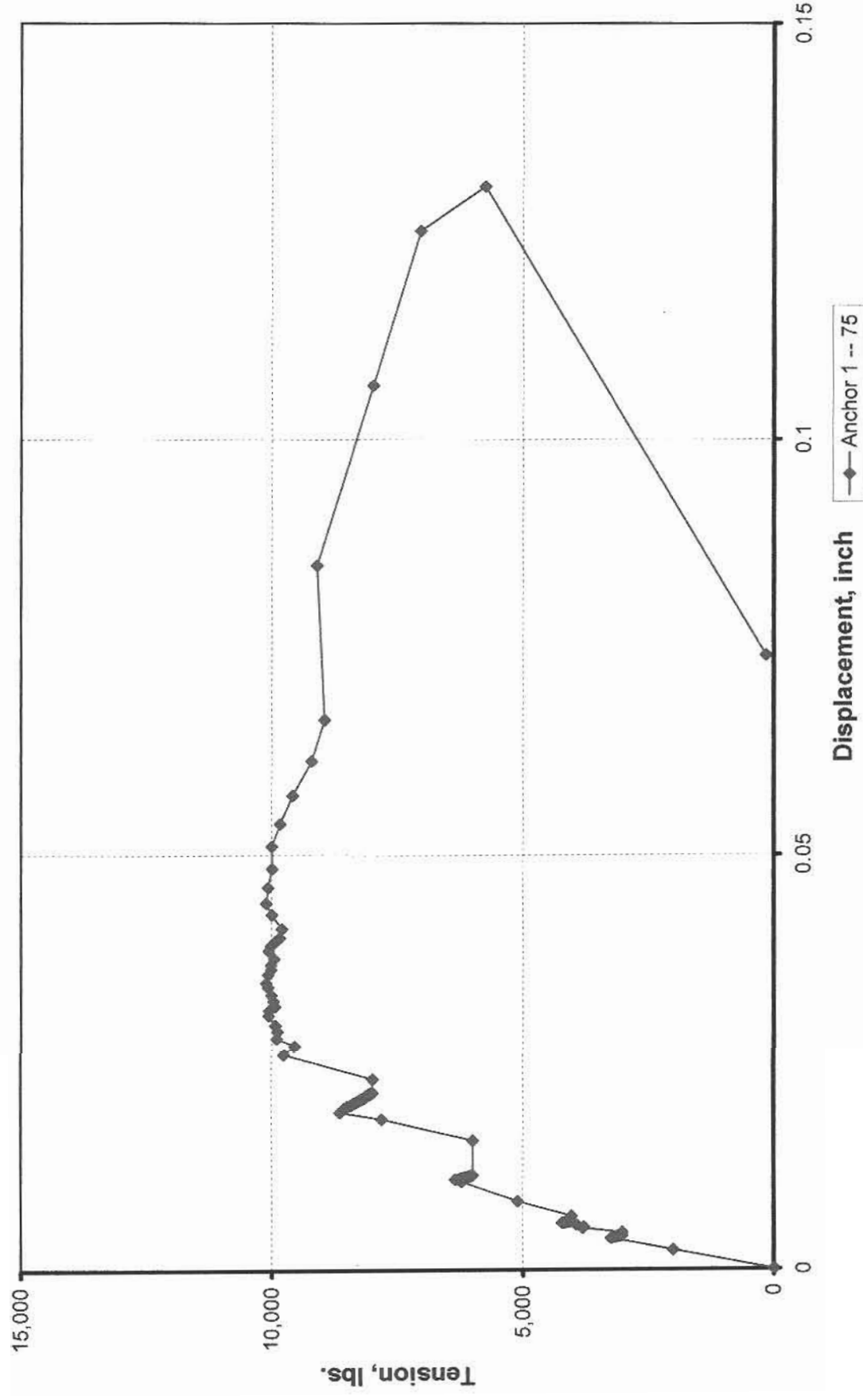
APPENDIX – Complete test results Summary and Graphs

Summary of Results

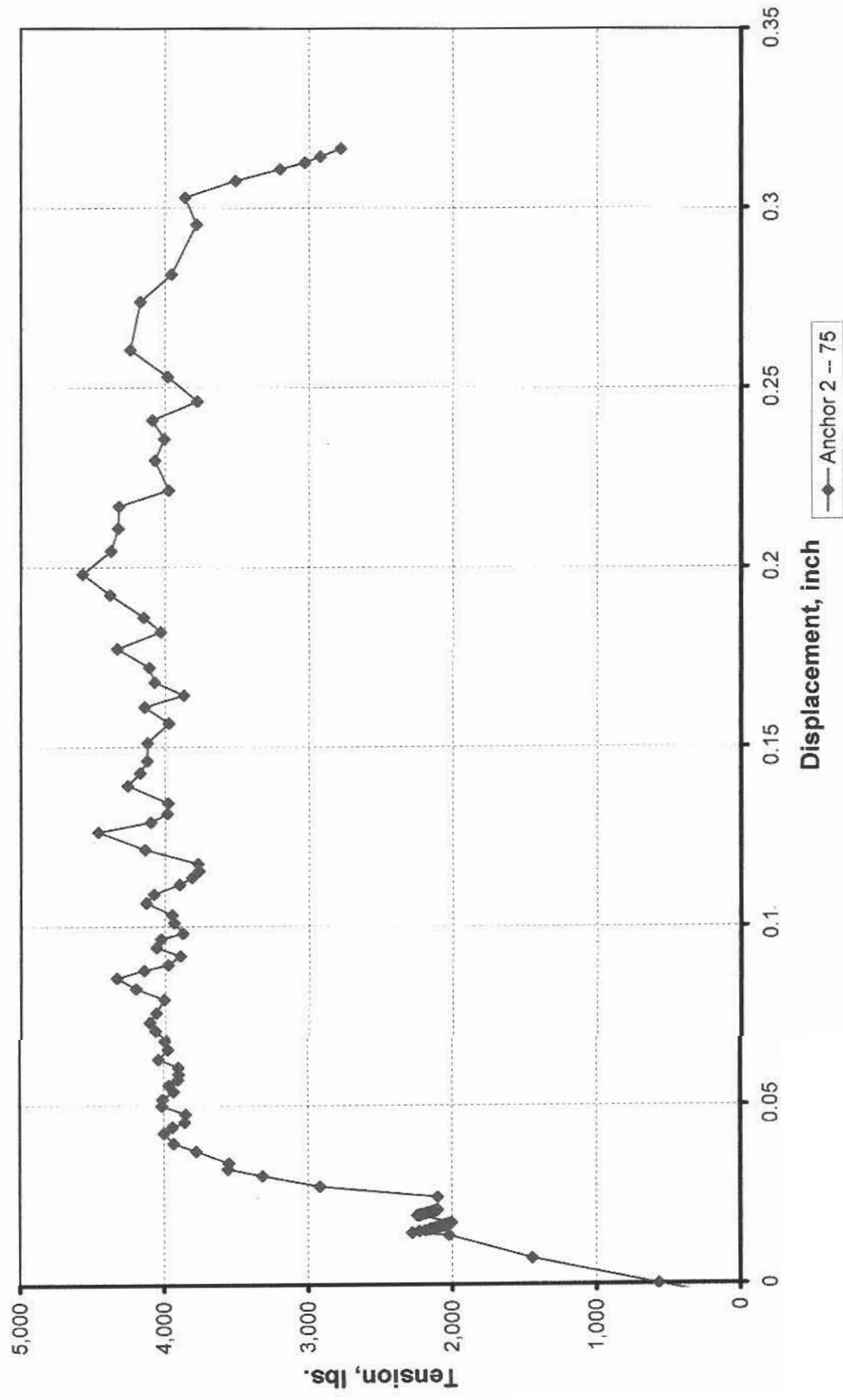
75 degree ultimate test results			110 degree ultimate test results		
anchor	failed at pounds		anchor	failed at inches	
1	10120.3		5	0.24	
2	4575.9		7	0.25	
4	6284.9		8	0.14	
6	7028.9		9	0.10	
11	18057.1		16	0.06	
Avg:	9213.4	times 85% = 7831.41	Avg:	0.16	times 115% = 0.18
14088.7 re-figured average			0.10 re-figured average		

Powers Fasteners bolt test for
NTSB by Standard Testing

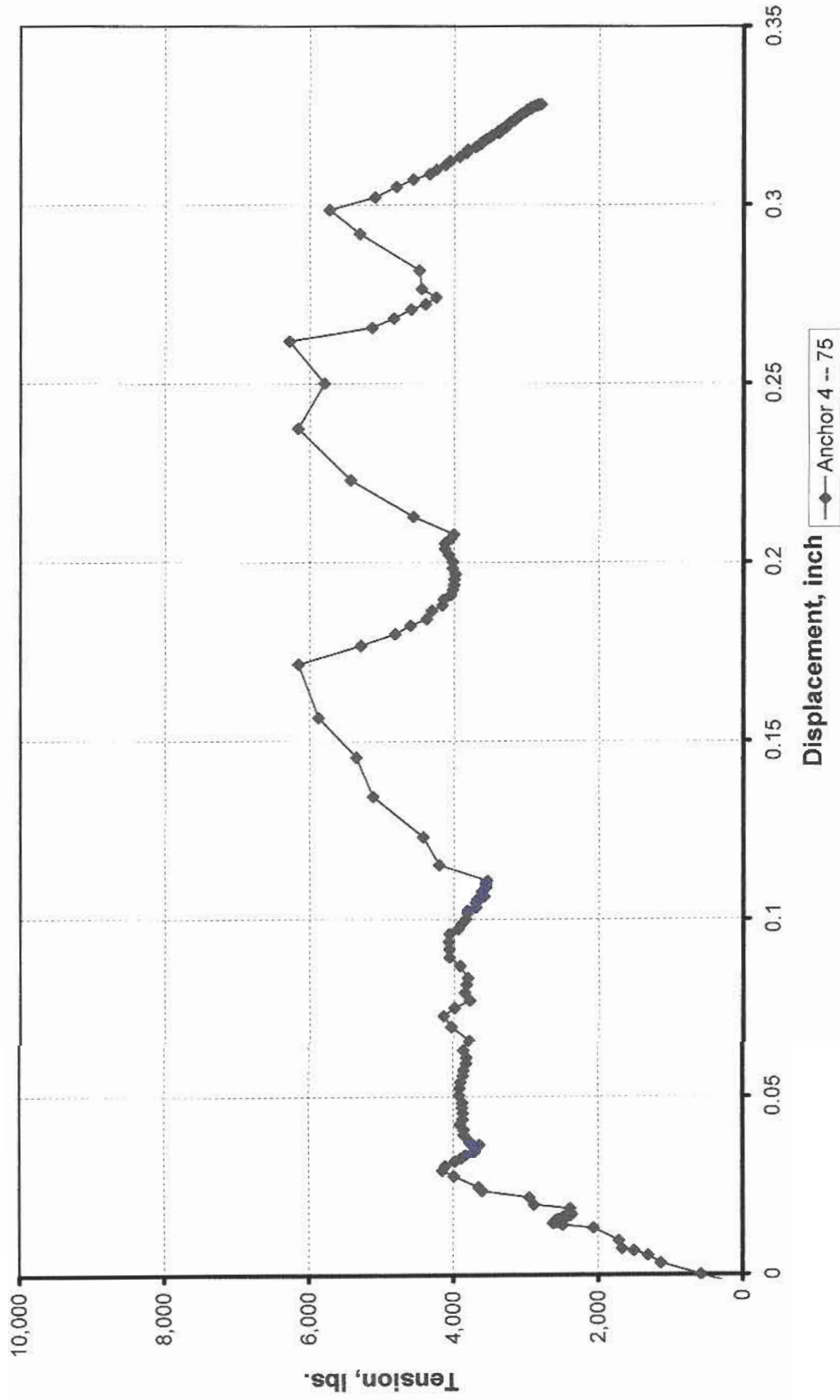
Anchor Tension Test ASTM D488



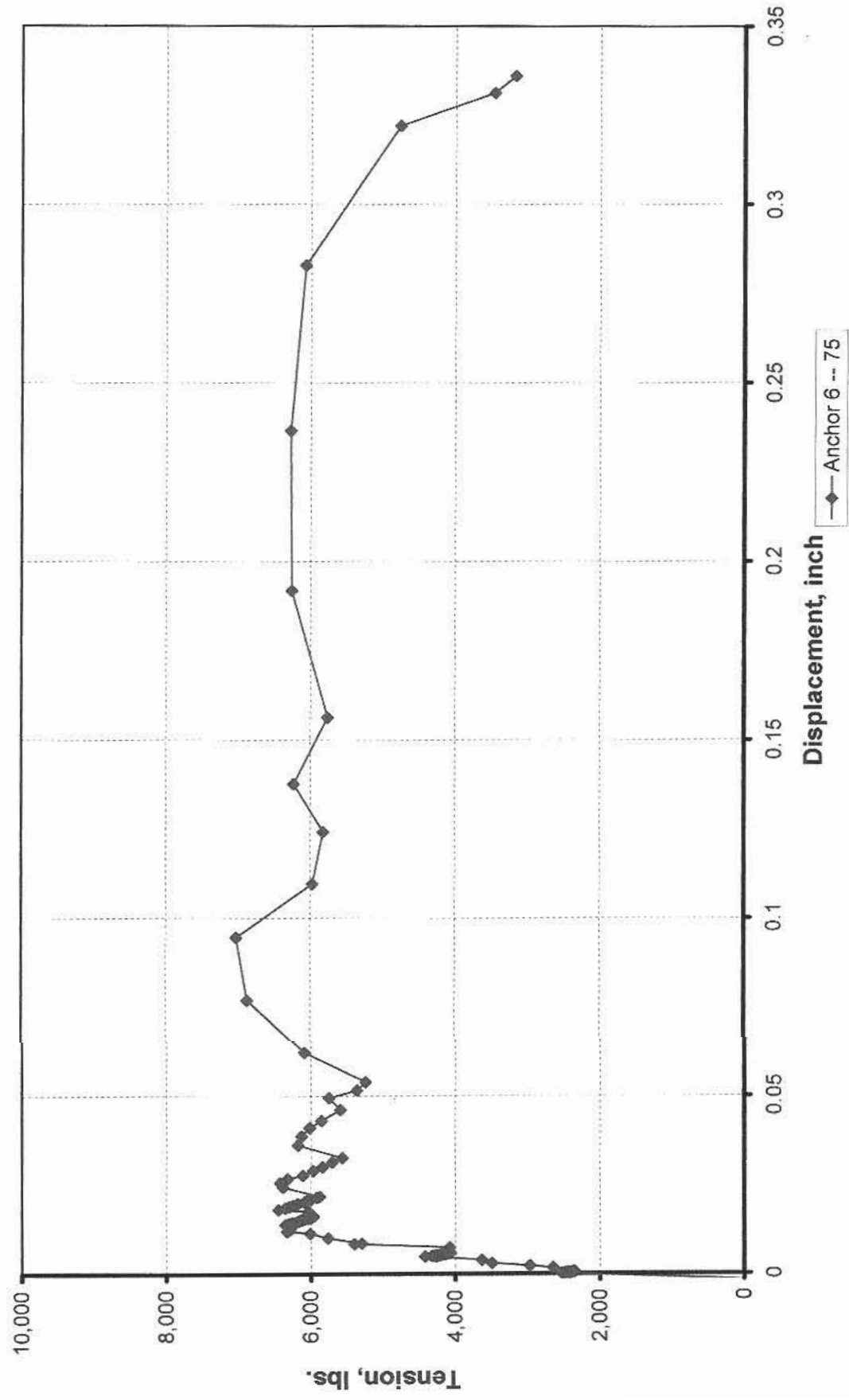
Anchor Tension Test ASTM D488



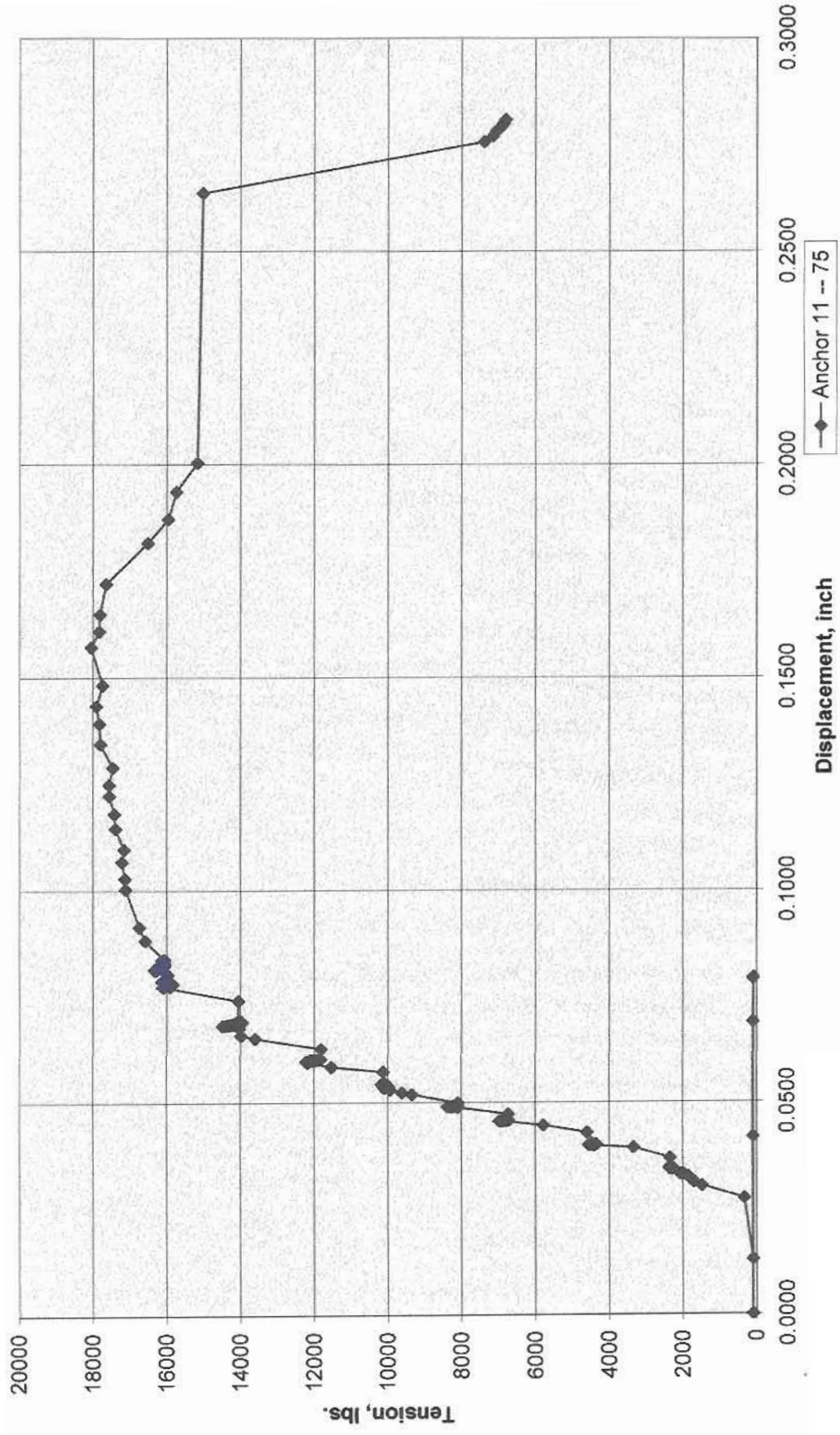
Anchor Tension Test ASTM D488



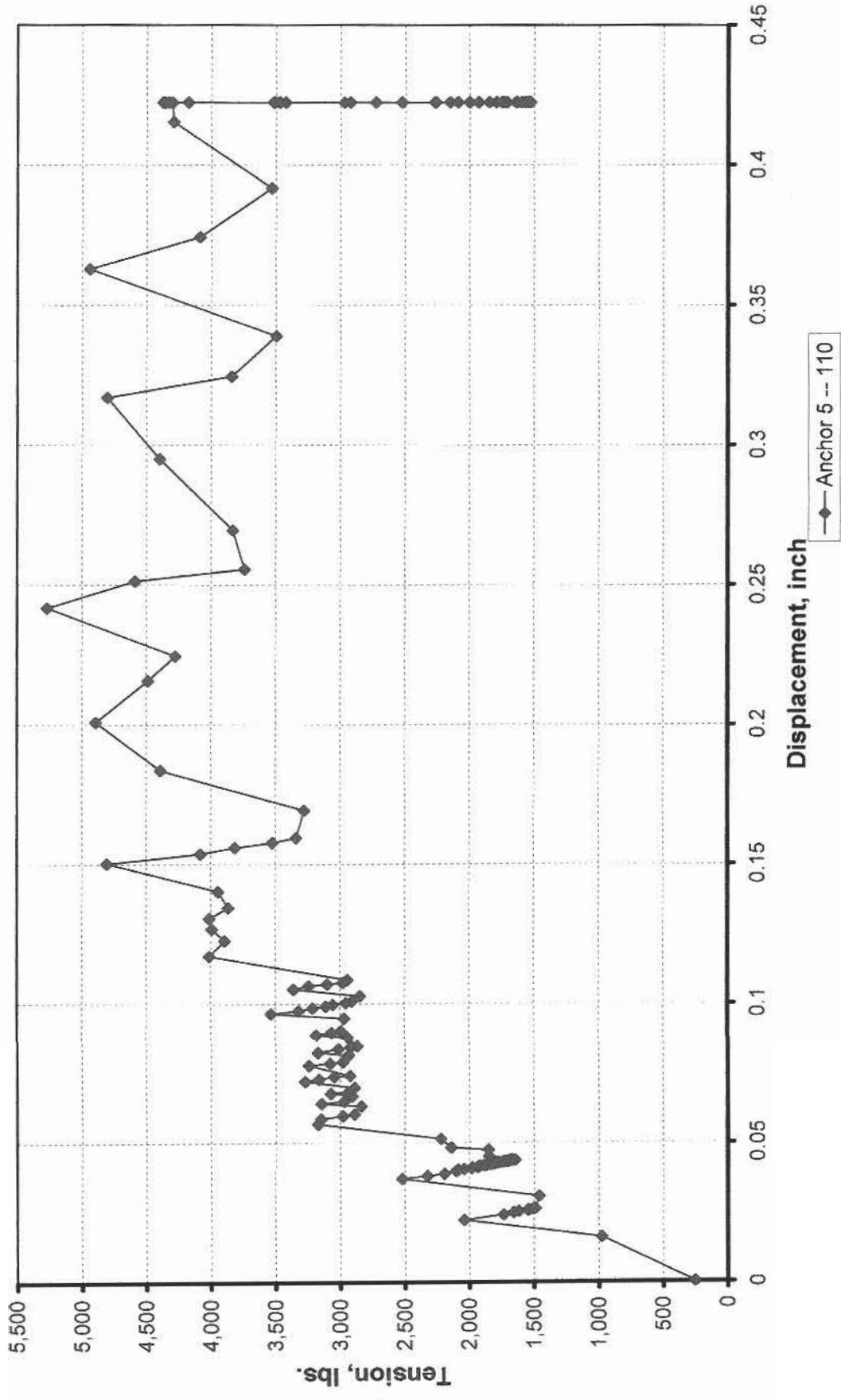
Anchor Tension Test ASTM D488



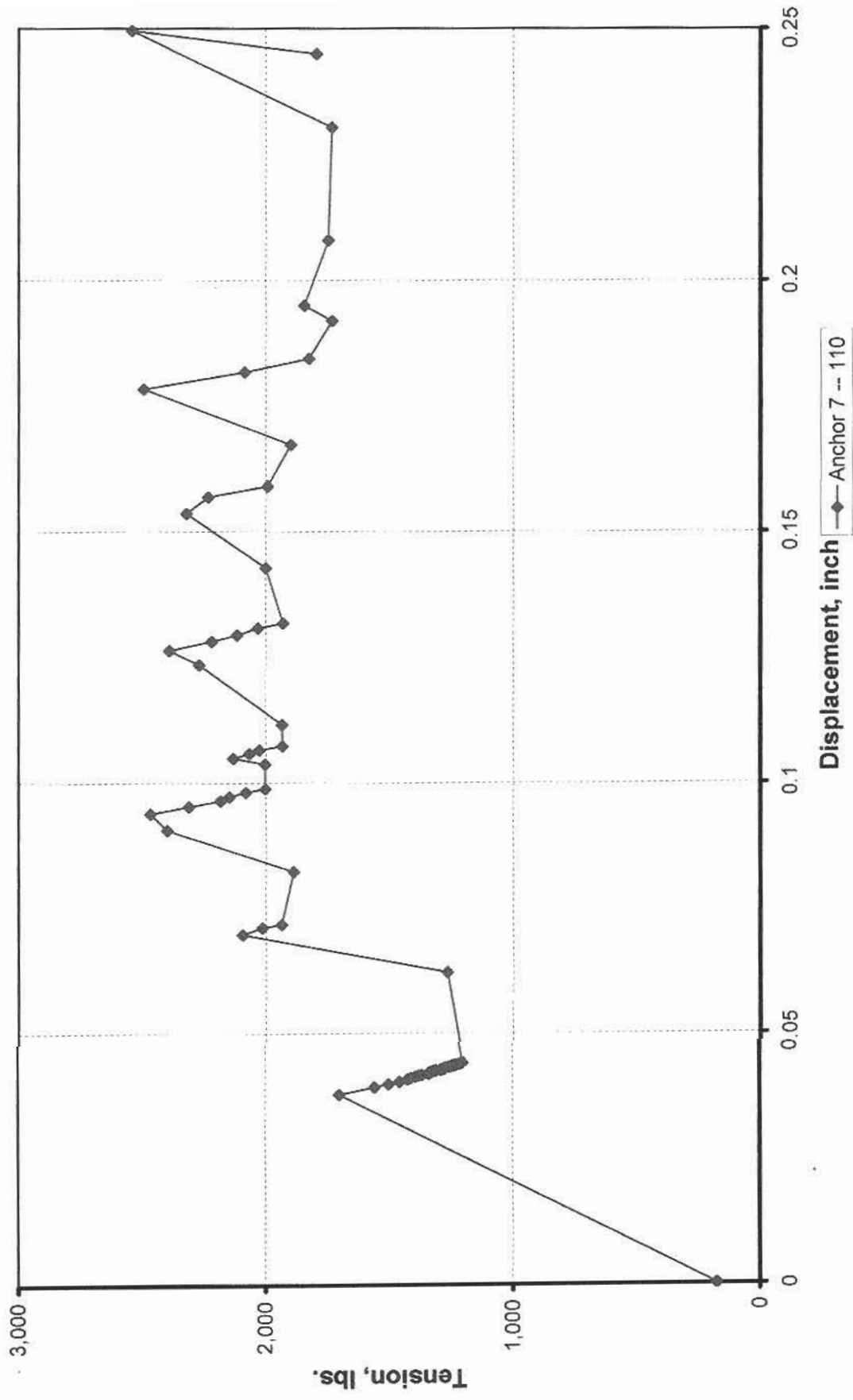
Anchor Tension Test ASTM D488



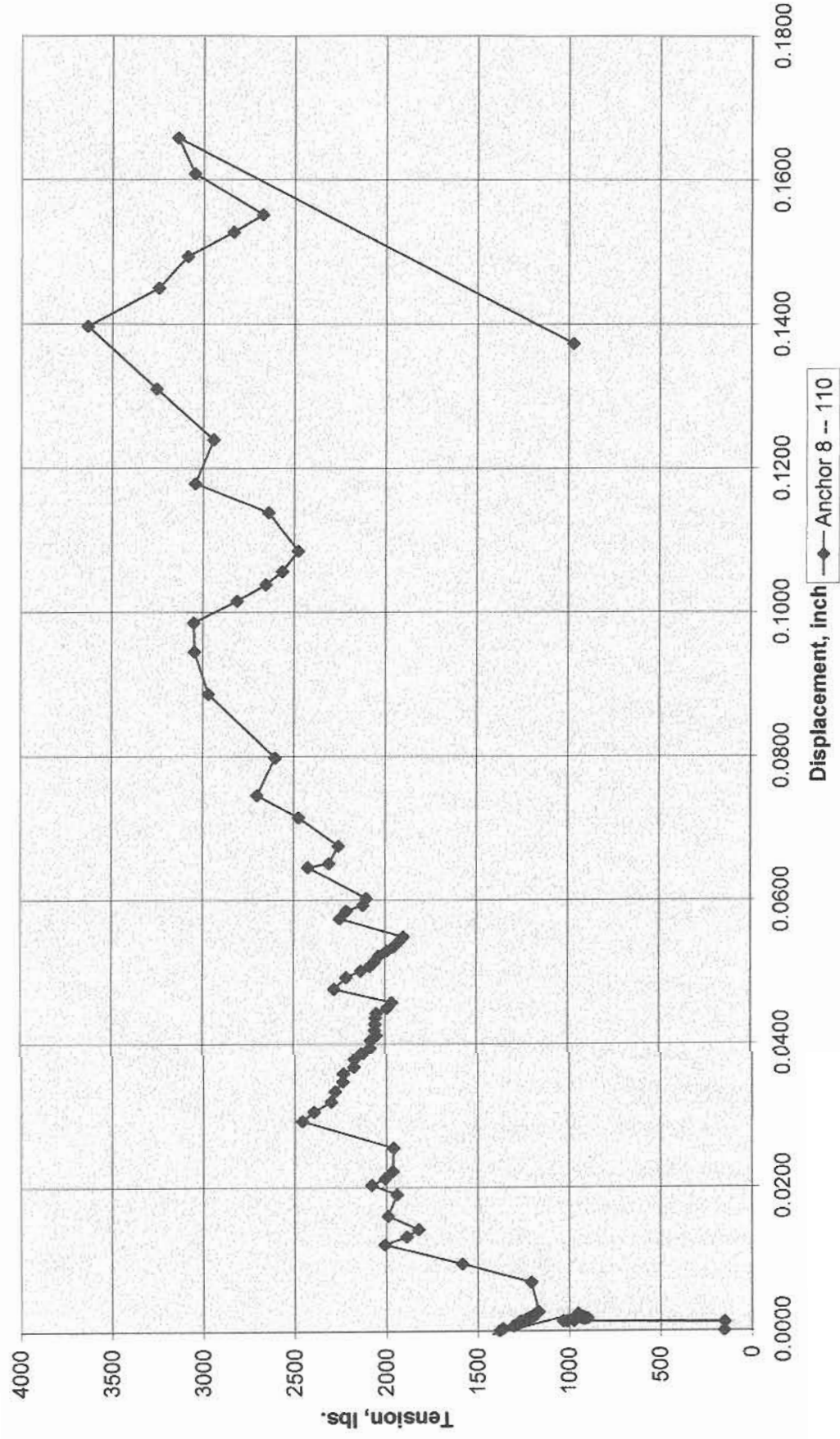
Anchor Tension Test ASTM D488



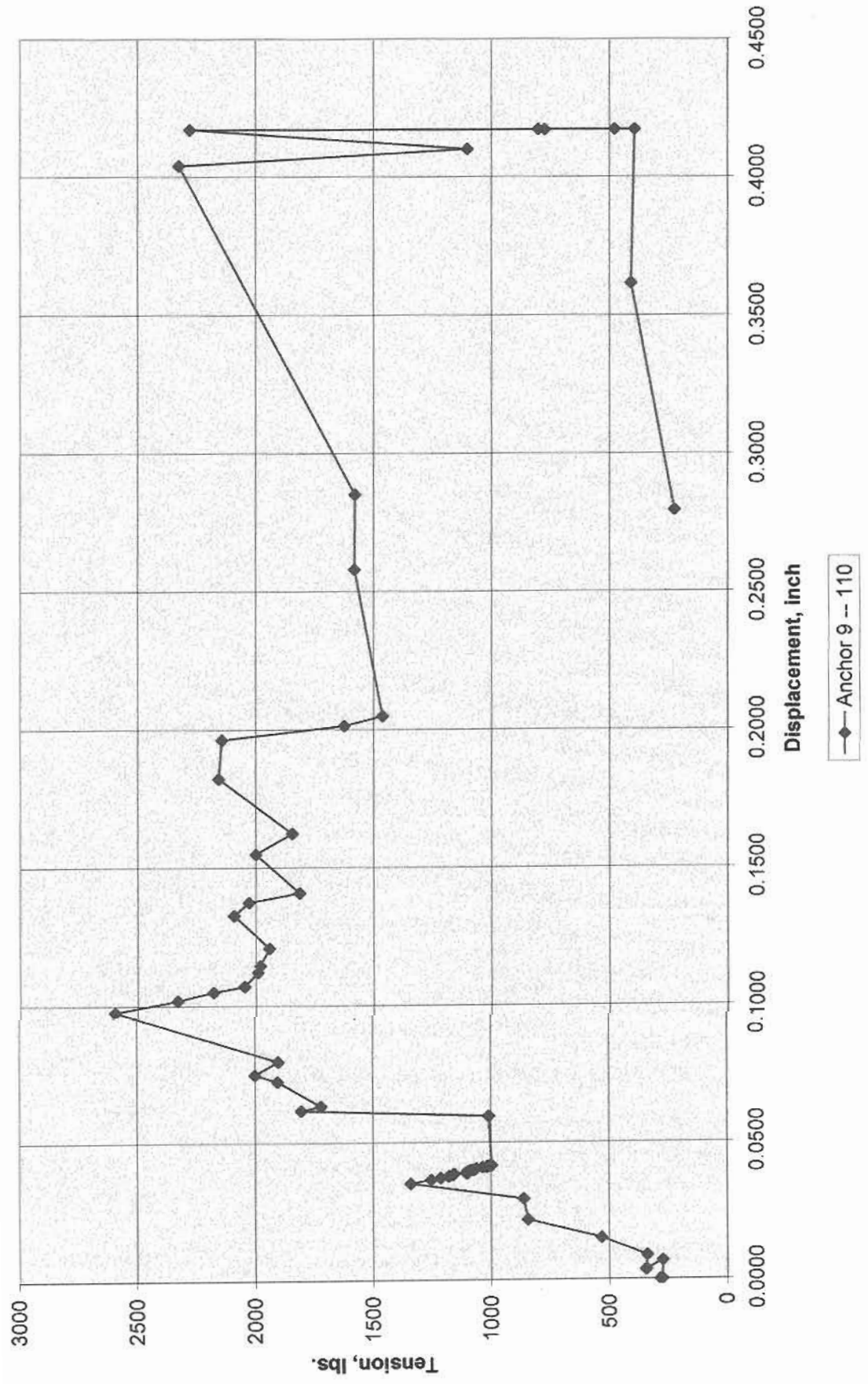
Anchor Tension Test ASTM D488



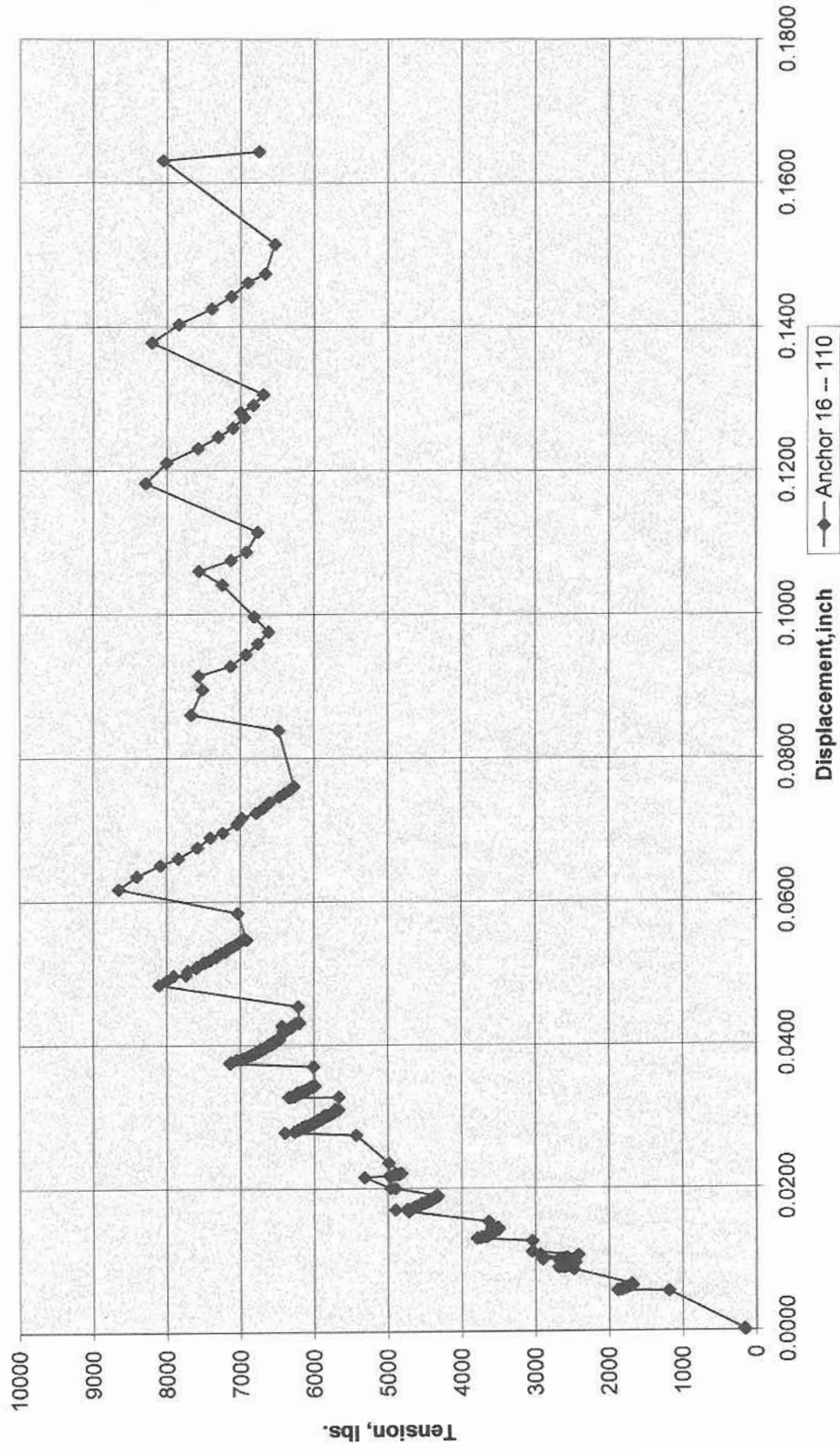
Anchor Tension Test ASTM D488



Anchor Tension Test ASTM D488



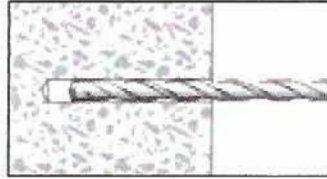
Anchor Tension Test ASTM D488



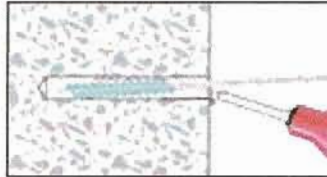
INSTALLATION GUIDELINES

Solid Base Materials

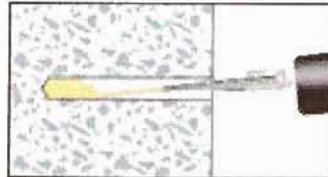
Drill a hole to the size and embedment required. The tolerances of the drill bit used should meet the requirements of ANSI Standard B212.15.



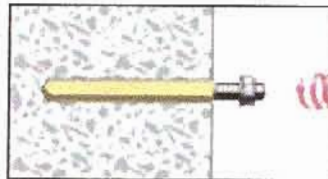
Starting from the bottom or back of the anchor hole, blow clean with compressed air for a minimum of four seconds, brush the hole with a nylon brush a minimum of four times, and blow it clean again. Vacuuming only is not sufficient. Blow out bulbs generally do not provide enough dust removal for most drilled anchor holes. Holes should be clean and sound. They may be dry or damp, but should be free of standing water or frost. If using reinforcing bar, be sure the bar will fit into the drilled hole. If a larger hole is required, the diameter should be as close as possible to the diameter of the reinforcing bar.



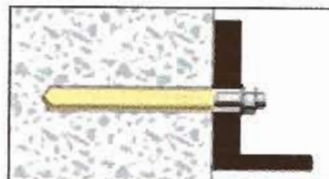
Prior to dispensing into anchor hole, balance the cartridge and visually inspect that adhesive components are uniformly mixed. Fill the hole approximately half to two-thirds full with adhesive starting from the bottom or back of the anchor hole. Slowly withdraw the static mixing nozzle as the hole fills to avoid creating air pockets.



Push the threaded rod or reinforcing bar into the anchor hole while turning slightly to ensure positive distribution of the adhesive. Be sure the rod is fully seated at the bottom of the hole and that some adhesive has flowed from the top of the hole. The threaded rod or reinforcing bar used should be free of dirt, grease, oil or other foreign material.

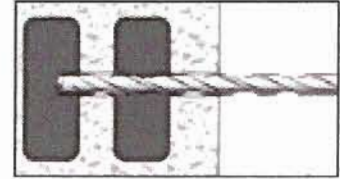


Allow the adhesive to cure for the specified time prior to applying any load. Do not disturb, torque, or load the anchor until it is fully cured.



Hollow Base Materials

Drill a hole to the size and embedment for the required screen size. The tolerances of the drill bit used should meet the requirements of ANSI B212.15.

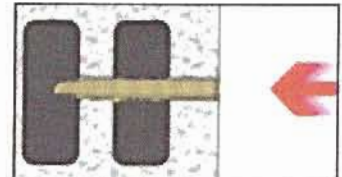


Starting from the bottom or back of the anchor hole, blow the hole clean with compressed air for a minimum of four seconds, brush the hole with a nylon brush a minimum of four times, and blow it clean again. Vacuuming only is not sufficient. Blow out bulbs generally do not provide enough dust removal for drilled anchor holes in hollow base materials.

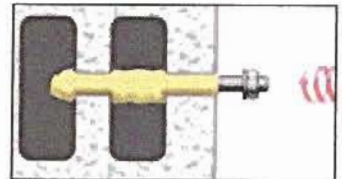
Prior to dispensing into screen tube, balance the cartridge and visually inspect that adhesive components are uniformly mixed. Fill the screen tube with adhesive starting from the bottom of the screen. The screen tube should be completely filled prior to inserting it in the anchor hole.



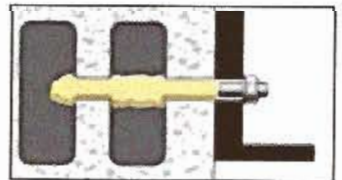
Insert the filled screen tube into the anchor hole until it is fully seated at the required embedment.



Push the threaded rod into the screen while turning slightly to ensure positive dispensing of the adhesive. Be sure the rod is fully inserted down to the end of the screen tube. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.



Allow the adhesive to cure for the specified time prior to applying any load. Do not disturb, torque, or load the anchor until it is fully cured.



Prior to use, read product label, Material Safety Data Sheet and injection tool instructions. Hole cleaning is essential for the proper performance of adhesive anchors.

For special applications including overhead installation, reference information on page 249.



TULSA OFFICE
5358 S 125 East Ave Ste B • Tulsa, OK 74146-6208
(918) 459-2700 • (800) 725-4592 • (918) 459-2715 FAX

902 Trails West Loop
Erid, OK 73703-6336
(580) 237-3130
(800) 725-3130
(580) 237-2311 FAX

202 SE "J" Ave
Lawton, OK 73501-2481
(580) 353-0872
(800) 725-0872
(580) 353-1263 FAX

3400 N Lincoln Blvd
Oklahoma City, OK 73105-5493
(405) 528-0541
(800) 725-0541
(405) 528-0559 FAX

May 21, 2007

National Transportation Safety Board
NTSB RE-30
490 L'Enfant Plaza East, SW
Washington, DC
20594

Attention: Carl Schultheisz
Materials Research Engineer

Subject: Correction to the Report on the Performance of Powers Fasteners' "High Strength Epoxy Anchoring System for Concrete and Masonry" Bolt Epoxy Power-Fast+ Fast Set Formula, dated May 16, 2007

Project Number: 2120-4008

Dear Mr. Schultheisz:

This letter is intended to correct a mistake in the report referenced above that I sent to you under separate cover, dated May 16, 2007.

There are some places in the report that refer to the 1/4" diameter bolts. The actual measurement is 1/2" diameter.

Respectfully submitted,
STANDARD TESTING AND ENGINEERING COMPANY

Randall Baker, P.E.
Senior Engineer

Miles Hunter, P.E.
Tulsa Operations Manager

(IAS listing #TL-187; Oklahoma CA#77 expires 6-30-07)